

Assignment 1: Containers with Docker

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1. BUILD & EXECUTION COMMANDS

```
make          # clean, build, and start both services
make down     # stop and remove the containers and volumes
```

Or without make:
docker compose up --build

2. STANDARD OUTPUT (STDOUT)

```
{
  "total_trips": 6,
  "avg_fare_by_city": [
    {
      "city": "Charlotte",
      "avg_fare": 16.25
    },
    {
      "city": "New York",
      "avg_fare": 19.0
    },
    {
      "city": "San Francisco",
      "avg_fare": 20.25
    }
  ],
  "top_by_minutes": [
    {
      "city": "San Francisco",
      "minutes": 28,
      "fare": 29.3
    },
    {
      "city": "New York",
      "minutes": 26,
      "fare": 27.1
    },
    {
      "city": "Charlotte",
      "minutes": 21,
      "fare": 20.0
    }
  ]
}
```

```
"city": "Charlotte",
"minutes": 12,
"fare": 12.5
},
{
"city": "San Francisco",
"minutes": 11,
"fare": 11.2
},
{
"city": "New York",
"minutes": 9,
"fare": 10.9
}
]
}
```

3. REFLECTION

This assignment showed how Docker Compose connects containers and uses healthchecks to coordinate startup. Writing the batch script reinforced the importance of proper data type conversion, especially when handling PostgreSQL numeric types, cursor tuples, and Python JSON serialization. It also demonstrated how to streamline the workflow with a Makefile, allowing anyone to build, start, and clean up the containers with a single command.